

Solution Architecture Specification

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. Architecture Diagram Overview

Decoupled 7-Tier architecture model separating the presentation client from backend pipeline processors and model storage caches.

2. Layer Definitions

- User Layer: Chrome/Firefox Web Browser clients.
- Presentation Layer: index.html CSS styles and main.js frontend controllers.
- Application Layer: Python Flask routing controller (app.py).
- Machine Learning Layer: Scikit-learn modeling engine (train_models.py).
- Data Layer: Numerical StandardScaler, categorical mappings, and High_Risk transforms.
- Model Storage: Saved model.pkl and preprocessing preprocessor.pkl.
- Deployment Layer: IBM Watson cloud hosting pipelines (watson_pipeline.py).